

Article

Four-Channel CIEL*a*b*-Infrared Image Representation for CNN-Based Oil Palm Fresh Fruit Bunch Ripeness Classification

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Abstract

Oil palm fresh fruit bunch (FFB) ripeness classification is essential for improving the oil extraction rate, oil quality, and mill processing efficiency. However, RGB-based classification is often limited by insufficient colour information from dark unripe fruitlets and shadowed regions of the bunch surface. This study evaluated the effect of image input representation on convolutional neural network (CNN)-based FFB ripeness classification. Four models with the same CNN architecture were compared using RGB, RGB + infrared (IR), CIEL*a*b*, and CIEL*a*b* + IR inputs. RGB and IR images were acquired using a dual-camera setup positioned 3 m from the FFB sample. The CIEL*a*b* + IR model achieved the best overall performance, with weighted-average precision, recall, and F1-score values of 0.84, 0.81, and 0.81, respectively, compared with 0.79, 0.76, and 0.76 for RGB-only. The addition of IR improved model performance by providing complementary near-infrared reflectance information, while CIEL*a*b* colour-space transformation provided a more discriminative colour representation. Class activation heat maps showed that the models focused mainly on fruitlet regions, with the CIEL*a*b* + IR model producing more distinct activation over ripeness-relevant areas. These findings demonstrate that the proposed four-channel CIEL*a*b* + IR image representation improves CNN-based FFB ripeness classification by combining lightness-separated colour information with near-infrared reflectance features, although underripe FFB remains challenging, where it produced the highest rate of false positives because of its transitional and heterogeneous characteristics.

Keywords: oil palm fresh fruit bunch (FFB); ripeness classification; convolutional neural network (CNN); CIEL*a*b* colour space; near-infrared (NIR) imaging; four-channel image representation; multimodal data fusion



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1. Introduction

Oil palm fresh fruit bunch (FFB) ripeness grading is a critical quality-control process in the palm oil industry. The ripeness condition of harvested FFB directly influences the oil extraction rate (OER), oil quality, free fatty acid (FFA) content, and overall mill processing efficiency [1–3]. Unripe or underripe bunches may reduce oil yield, whereas overripe bunches may increase the risk of oil deterioration, including the formation of

free fatty acids [1]. Accurate and consistent FFB ripeness classification is therefore essential for improving harvesting decisions, mill grading operations, and overall production quality [2,4,5].

Despite its importance, FFB ripeness grading remains challenging. Conventional grading is commonly based on visual assessment of fruitlet colour and bunch appearance. However, such assessment is subjective and can be affected by lighting conditions, shadow, fruitlet arrangement, and variation in bunch morphology [2,4]. Outdoor image acquisition introduces additional challenges arising from natural illumination, background complexity, class imbalance, and variations in FFB appearance [6,7]. The central problem addressed in this study is the limited and non-uniform colour information available from RGB images of FFB, particularly for dark unripe fruitlets and fruitlets located in shadowed regions [1,8]. These factors can reduce the visibility of ripeness-related features and make CNN-based classification less reliable.

RGB imaging is widely used for agricultural product classification because it is simple, low-cost, and capable of capturing visible colour and texture information [8,9]. Previous FFB studies have also demonstrated that colour, texture, and their combinations can provide useful discriminatory information for maturity classification using machine-learning and image-processing approaches [10–12]. However, RGB images are illumination-dependent and may not fully capture discriminative features from dark or shadowed regions of complex biological structures [13]. This limitation is particularly evident in FFB, where densely packed fruitlets create self-shadowing and uneven illumination across the bunch surface. Furthermore, RGB pixel intensities are linked to ripeness-related pigment changes, including chlorophyll degradation and changes in red colour response, but these colour-based features can still be affected by illumination variation and non-fruit artefacts [14]. Alternative colour representations have therefore also been explored for FFB grading, including colour-space-based approaches intended to improve the separation of ripeness-related colour information [15,16].

Infrared (IR) imaging provides additional information beyond the visible colour spectrum. Unlike RGB imaging, IR imaging captures reflectance characteristics that are less dependent on visible colour appearance and can therefore provide complementary information for agricultural quality inspection and fruit maturity evaluation [4,17]. Other non-visible and active sensing approaches, including thermal imaging and LiDAR-based reflectance measurements, have similarly demonstrated that maturity-related physical characteristics of FFB can be detected beyond conventional RGB colour information [12,15]. In the context of oil palm FFB, previous laser-based reflectance studies have shown that visible and near-infrared wavelengths are sensitive to ripeness-related changes in FFB reflectance [18]. Recent reviews of FFB ripeness sensing have likewise highlighted the potential of combining complementary optical and sensing modalities to improve ripeness assessment [5].

In agricultural image analysis, RGB and NIR data have been combined as four-channel image inputs to exploit visible colour information and near-infrared spectral responses for improved deep learning-based interpretation of crop scenes [10]. Similar multimodal RGB-NIR strategies have also improved fruit detection and agricultural product inspection, indicating that NIR information provides complementary features for agricultural imaging tasks [19,20].

During fruit ripening, changes in chlorophyll content, moisture, tissue structure, and pigment composition influence how near-infrared light is absorbed and scattered by fruit tissues. These changes can alter the IR reflectance of the fruitlet surface, particularly in the lower near-infrared region, where ripeness-dependent spectral differences are more evident than visible colour differences under inconsistent illumination and shadowed

conditions [4,17,18]. Non-destructive optical studies of FFB have also associated spectral responses with changes in moisture and oil-related properties during fruit development [3]. Therefore, IR imaging can reveal ripeness-related characteristics that may not be clearly captured by RGB images alone, especially in shadowed regions or on dark-coloured fruitlets.

Colour-space transformation may also improve image-based classification. The CIEL*a*b* colour space is attractive for agricultural product grading because it separates lightness from chromatic components, providing a more perceptually uniform representation of colour [8,13]. Studies using alternative colour representations for FFB maturity assessment have similarly shown that colour-space transformation can provide useful ripeness-related information [21,22]. By separating lightness from chromatic information, CIEL*a*b* may reduce the influence of inconsistent illumination while preserving colour differences associated with ripening. In the CIEL*a*b* colour space, brightness is represented by the L* channel, while colour information is separated into the a* (green-to-red) and b* (blue-to-yellow) channels. As oil palm FFB ripens, the fruitlets gradually change in colour from darker greenish tones to reddish-orange tones. Separating brightness from colour information allows the CNN to better distinguish ripeness-related colour variations while minimizing the influence of uneven lighting conditions. Consequently, CIEL*a*b* images may provide a more discriminative colour representation than conventional RGB images for FFB ripeness classification.

Convolutional neural networks (CNNs) have enabled automatic feature extraction from image data and have shown strong potential for agricultural classification tasks [9,23,24]. CNN-based FFB classification has subsequently been explored using mobile deep learning systems, visual-attention mechanisms, real-time object detection frameworks, lightweight networks, and transfer-learning approaches, demonstrating the growing potential of deep learning for automated FFB grading [16,25–28]. However, CNN performance depends not only on network architecture but also on the quality and representation of the input image data. For FFB ripeness classification, RGB images alone may limit model performance because of restricted colour information in dark fruitlets and shadowed regions. Incorporating IR imaging and improved colour-space representation may therefore provide richer and more robust input features for CNN-based classification.

Previous studies have investigated RGB imaging, multispectral imaging and infrared sensing, colour-space transformation, traditional machine-learning methods, CNN-based approaches, and multimodal deep learning strategies for agricultural and oil palm fruit assessment [2–7,10–12,14–18,21,22,25–29]. These studies demonstrate continuing advances in colour-texture analysis, alternative colour spaces, non-visible sensing, lightweight deep learning, real-time detection, and complementary sensing approaches for agricultural assessment. However, comparative studies that evaluate these image representations using an identical CNN architecture for FFB ripeness classification remain limited. It therefore remains difficult to determine whether performance improvements are due to the image representation itself or differences in network design.

This study investigates the effect of image input representation on CNN-based FFB ripeness classification. Four input formats were evaluated: RGB, RGB + IR, CIEL*a*b*, and CIEL*a*b* + IR. The same CNN architecture was used for all models to isolate the effect of input image representation on ripeness classification performance. Among the evaluated input formats, the proposed four-channel CIEL*a*b* + IR representation combines complementary colour and near-infrared information, providing a more comprehensive image representation for FFB ripeness classification. The main innovations and core contributions of this work are summarized as follows:

- **Novel Four-Channel Image Representation:** A unified four-channel CIEL*a*b* + IR input format (L^* , a^* , b^* and a single 700–1100 nm IR channel) as complementary chromatic and IR spectral data to a deep learning pipeline is proposed.
- **Controlled Experimental Isolation:** By employing an identical lightweight CNN architecture across four input formats (RGB, RGB+IR, CIEL*a*b**, and CIEL*a*b** + IR), the performance gains attributable solely to the input image representation can be isolated and quantified, independent of variations in network complexity.
- **Spatial Interpretability via Class Activation Mapping:** Grad-CAM analysis visually demonstrates that the proposed CIEL*a*b** + IR representation enables the network to concentrate its attention specifically on biologically relevant fruitlet surface regions rather than on non-informative background or shadow regions.

2. Materials and Methods

2.1. Image Acquisition Setup

Image acquisition was conducted using two synchronized CMOS cameras (1/2" sensor format, 1280×1024 native pixel resolution): one dedicated RGB camera (model: Hikrobot MV-CU013-A0GC) and one monochrome Infrared (IR) camera (model: Hikrobot MV-CU013-A0GM, equipped with an optical bandpass filter in the range of 700–1100 nm). This wavelength range includes the red-edge region, where chlorophyll absorption changes significantly as fruit ripens, providing additional information that cannot be fully captured by visible RGB images alone [14,19]), both camera manufacturers are based in Hangzhou, China. The two cameras were mounted side-by-side with a parallel baseline distance of approximately 8 cm. The acquisition distance measured from the outermost edge of the camera lenses to the fresh fruit bunch (FFB) was maintained at 3 m, yielding a horizontal field of view (FOV) of approximately 11.4° and a focal length of about 32 mm, which covers typical FFB dimensions of about $45 \text{ cm} \times 36 \text{ cm}$ (Figure 1). Dual-camera capture was controlled simultaneously via custom software built on the camera manufacturer's Software Development Kit (SDK) v4.4.1.3, maintaining inter-sensor capture latency below 1.0 s.

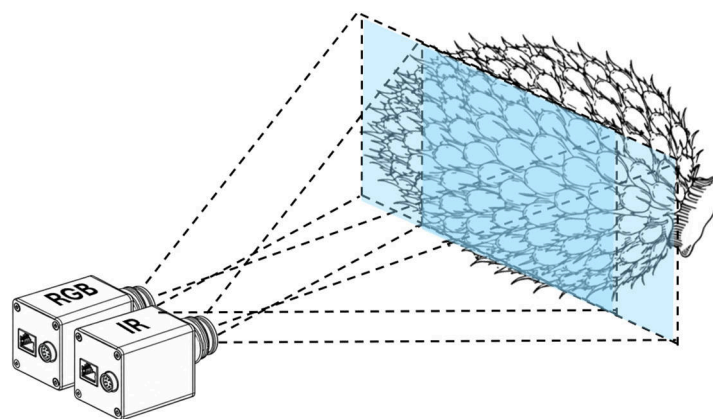


Figure 1. Dual-camera imaging setup for FFB image acquisition. The RGB and IR cameras were arranged in parallel and directed toward the same FFB sample at a fixed distance of 3 m.

Pre-acquisition geometric calibration was performed to ensure a large overlapping FOV in the target working space. Using spatial features from the calibration targets, manual pixel-level image registration was executed to align the IR channel with the RGB frame.

2.2. CNN Model Development

CNN models were developed to evaluate the effect of image input representation on FFB ripeness classification. The evaluated input formats were RGB, CIEL*a*b*, RGB + IR, and CIEL*a*b* + IR. To ensure a fair comparison, the same CNN classification pipeline was used for all models; only the input image format was varied. The CNN architecture comprised convolutional and max-pooling layers for feature extraction, followed by fully connected layers, a dropout layer, and an output layer for ripeness classification [23].

For the RGB-only model, the original three-channel RGB image was used as input. For the CIEL*a*b* model, each RGB image was converted into the CIEL*a*b* colour space. The L* (lightness), a* (green-red), and b* (blue-yellow) channels were then used as the input to the CNN. For the RGB + IR and CIEL*a*b* + IR models, the corresponding IR image was combined with the colour-based input to provide additional spectral information. The CIEL*a*b* + IR model combined the three CIEL*a*b* channels, namely L* (lightness), a* (green-red), and b* (blue-yellow), with one corresponding IR channel to form a four-channel input for the CNN. A total of 279 paired (50 Overripe, 65 Ripe, 84 Underripe and 80 Unripe) RGB-IR images were acquired for the study. Prior to model training, the ripeness stage of each FFB was determined through visual assessment by two experienced oil palm fruit graders from the Malaysian Palm Oil Board (MPOB) based on visual ripeness characteristics, including fruitlet colour, fruitlet detachment, overall bunch appearance and mesocarp color, in accordance with standard harvesting practices.

2.3. Model Architecture

For models combining color and infrared data, early fusion is performed at the input layer. The 3-channel color tensor (RGB or CIEL*a*b*) and the 1-channel IR tensor are concatenated along the channel axis to create a single 4-channel input tensor ($4 \times 256 \times 256$). This fused tensor is fed directly into the first convolutional layer (Conv2D, 4 input channels), allowing the network to learn joint spatial-spectral features from the first layer onward.

To ensure complete reproducibility, Figure 2 outlines the layer-by-layer architectural design of the custom RipenessCNN model. The network comprises three feature extraction convolutional blocks. Each block integrates a 3×3 2D Convolutional layer, 2D Batch Normalization, a Rectified Linear Unit (*ReLU*) activation function, and a 2×2 Max-Pooling layer with a stride of 2. Spatial feature map dimensions are systematically reduced while channel capacity increases from 32 to 64 and 128.

Feature maps are aggregated via Adaptive Global Average Pooling into a $128 \times 1 \times 1$ tensor and flattened into a 128-dimensional vector. The classification head consists of a 64-unit fully connected layer regularized by Dropout ($p = 0.5$) and a final 4-unit linear output layer.

All models shared the same CNN architecture to ensure a fair comparison among the evaluated image representations. Using a common architecture allowed any differences in classification performance to be attributed to the input image representation rather than variations in network design.

The shared CNN pipeline consisted of convolutional layers, max-pooling layers, and fully connected layers. The convolutional layers extracted local spatial and textural features, such as fruitlet colour distribution, bunch surface texture, and ripeness-related visual patterns. The max-pooling layers reduced the spatial dimensions of the feature maps while preserving important features. After feature extraction, the learned feature maps were flattened and passed through fully connected layers. Dropout was applied to reduce overfitting and improve model generalisation. The final output layer produced the FFB ripeness classification result.

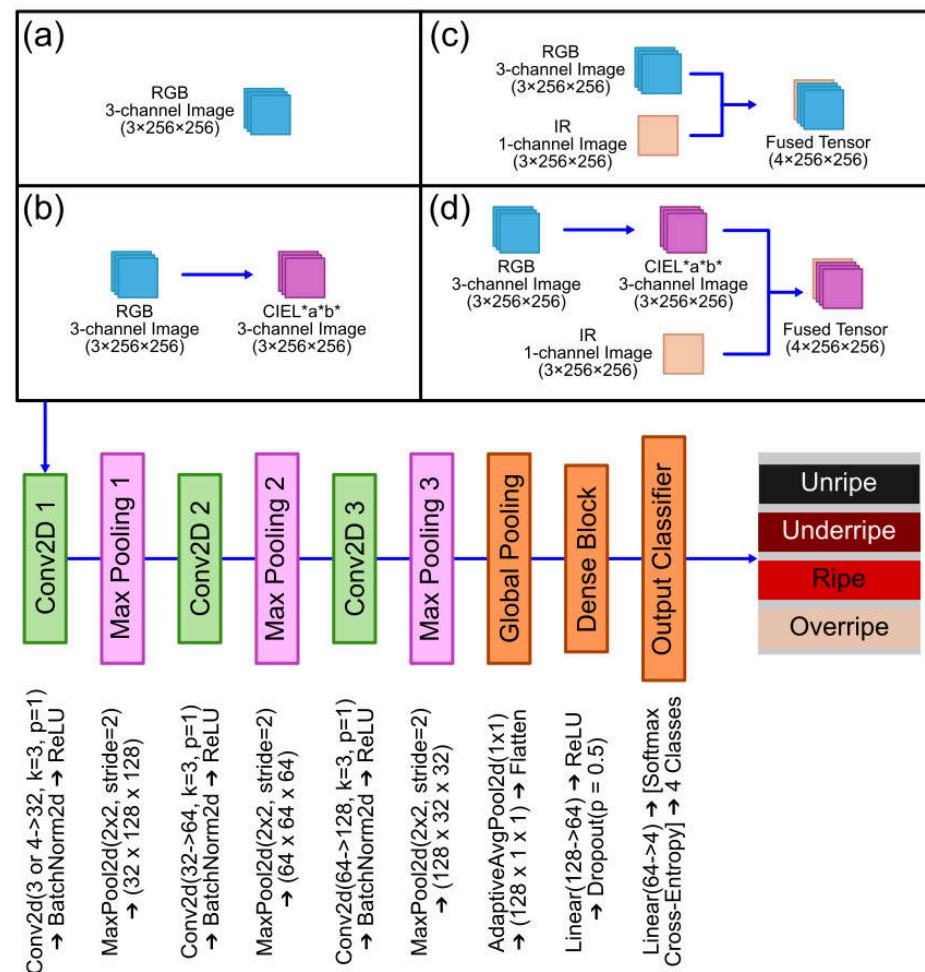


Figure 2. CNN architectures used for FFB ripeness classification with different input image formats: (a) RGB, (b) CIEL*a*b*, (c) RGB + IR, and (d) CIEL*a*b* + IR. All models used the same CNN classification pipeline and differed only in input image representation, where k is kernel size and p is padding.

Model performance across the four ripeness classes (Unripe, Underripe, Ripe, Overripe) was evaluated using macro-averaged Precision, Recall, and F_1 -score, as defined below:

$$Precision = \frac{TP}{(TP + FP)} \quad (1)$$

$$Recall = \frac{TP}{(TP + FN)} \quad (2)$$

$$F1 = 2 \times \frac{(Precision \times Recall)}{(Precision + Recall)} \quad (3)$$

where TP , FP , and FN denote the total counts of True Positives, False Positives, and False Negatives accumulated across evaluation iterations.

2.4. Training Configuration

The model was optimized using the Adam algorithm with a base learning rate of 0.001 and trained for a maximum of 100 epochs using a mini-batch size of 64. Dropout with a probability of 0.5 was applied to dense hidden connections to reduce overfitting, and two-dimensional mini-batch normalization was used after each convolutional layer. Early stopping monitored validation loss with a patience of 15 epochs. The dataset was

partitioned using a stratified hold-out protocol, with 80% of samples used for training and 20% used for independent testing. The four target classes were overripe, ripe, underripe, and unripe. Data augmentation included random horizontal flipping, random vertical flipping, and random spatial rotation within ± 15 degrees. To minimise the effect of random data partitioning and improve the reliability of model evaluation, the dataset was assessed using 10 independent random (note that 10-split does not equal 10 independent data points) train–test splits with a consistent 80:20 ratio. The support for the Overripe, Ripe, Underripe, and Unripe classes is 10, 13, 17, and 16, respectively. The reported performance metrics and corresponding standard deviations represent the average across these 10 runs, providing an assessment of both the classification performance and the consistency of each model. The training configuration and hyperparameters are summarized in Table 1.

Table 1. Summary of hyperparameters and training settings.

Hyperparameter	Configuration Details
Optimization algorithm	Adam
Base learning rate	0.001
Loss function	Categorical cross-entropy loss
Mini-batch size	64
Maximum epochs	100 (early stopping triggered before this limit when validation loss did not improve)
Regularization	Dropout probability = 0.5 for dense hidden connections
Feature standardization	Two-dimensional mini-batch normalization after each convolutional layer
Early stopping	Validation loss monitored with patience = 15 epochs
Model recovery	Weights restored from the minimum validation-loss configuration
Data partitioning	80% training and 20% independent testing
Target classes	Overripe, ripe, underripe, and unripe
Data augmentation	Random horizontal flip, random vertical flip, and random rotation within ± 15 degrees
Trainable parameters (RGB/ CIEL*a*b*)	82,852
Trainable parameters (RGB/CIEL*a*b* + IR)	83,140

All deep learning pipeline routines were implemented using Python 3.12 and PyTorch 2.5.1 compiled with CUDA 12.1. Model training and evaluation were executed on a localized workstation equipped with an NVIDIA GeForce RTX 4050 Laptop GPU (6.0 GB VRAM) and 16 GB System RAM.

3. Results and Discussion

3.1. Spectral Characteristics of Oil Palm Fruitlets

Figure 3 shows clear spectral differences among unripe, underripe, and ripe fruitlets, particularly in the lower near-infrared region of approximately 700–780 nm, where separation among the ripeness stages is evident. This region corresponds to the red-edge transition, where chlorophyll absorption decreases during ripening. Consequently, differences in reflectance become more apparent, allowing the ripeness stages to be distinguished more clearly.

The observed spectral variation is associated with physical changes that occur as fruit matures, including changes in chlorophyll content, moisture content, tissue structure, and pigment composition. These changes influence how near-infrared light is absorbed, scattered, and reflected from the fruitlet surface. Similar reflectance-based sensitivity to oil palm FFB ripeness has been demonstrated using IR-assisted laser remote sensing, where selected visible-NIR wavelengths were used to distinguish FFB ripeness categories and support non-destructive ripeness assessment [19].

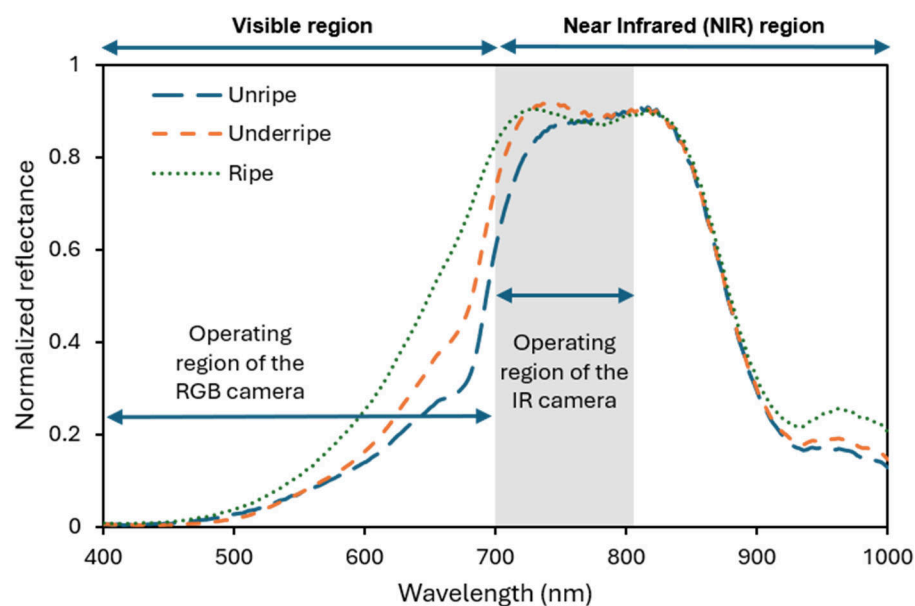


Figure 3. Average spectral reflectance of unripe, underripe, and ripe oil palm fruitlets across the visible and near-infrared spectrum. The shaded region indicates the operating wavelength range of the IR camera used for image acquisition.

The spectral characteristics in Figure 3 provide a physical justification for incorporating IR imaging into the proposed CNN classification framework. Unlike visible colour, which is strongly influenced by lighting and shadowing, near-infrared reflectance captures ripeness-related optical responses that can complement RGB information. By capturing reflectance information in the wavelength region where ripeness-related differences are prominent, the IR camera provides additional features that enhance the representation of FFB characteristics, particularly when combined with CIEL*a*b* colour representation.

3.2. Representative RGB and IR Images

Figure 4 shows representative FFB samples at four ripeness stages: unripe, underripe, ripe, and overripe. The RGB images show visible colour differences among ripeness stages. Unripe and underripe bunches generally exhibit darker and less uniformly developed fruitlet colouration, while ripe and overripe samples show more reddish-orange fruitlets and more pronounced colour variation across the bunch surface.

The dependence of RGB pixel intensity on pigment-related colour changes has previously been used to guide physics-informed ripeness classification, indicating that colour indices can capture ripening trends associated with fruit pigment changes [14]. However, RGB images may still be affected by shadow, dark fruitlets, specular reflection, and non-fruit structures. In contrast, the IR images provide complementary information that is not directly visible in the RGB images. While RGB images mainly represent visible colour and texture, IR images emphasize reflectance intensity differences across the bunch surface. These intensity variations may be related to fruitlet surface properties, moisture condition, physiological changes, or structural differences associated with ripeness development.

Together, the RGB and IR images indicate that FFB ripeness classification can benefit from both visible colour characteristics and non-visible spectral reflectance features.

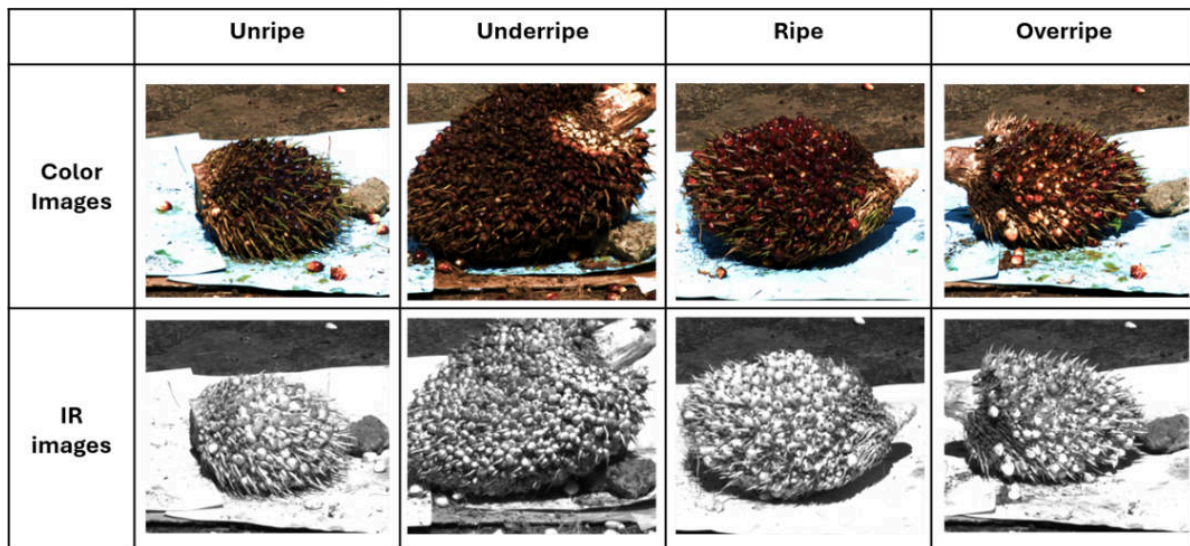


Figure 4. Representative RGB and infrared images of fresh fruit bunches at different ripeness stages. The columns show unripe, underripe, ripe, and overripe FFB samples from left to right. The (**upper**) row shows RGB images, while the (**lower**) row shows corresponding IR images.

3.3. Class Activation Heat-Map Analysis

Figure 5 presents class activation heat maps generated from the trained CNN models using the four input formats. The heat maps are overlaid on the FFB images to visualize image regions that contributed most strongly to the classification decision. For the multi-modal models (RGB + IR and CIEL*a*b* + IR), the activation maps were generated after feature fusion in the shared CNN backbone, representing the combined contribution of both colour and infrared information to the final classification. Across the models, activation regions were concentrated mainly on fruitlet areas rather than the surrounding and background, indicating that the CNNs learned to focus on biologically relevant ripeness features, including fruitlet colours, surface textures, and bunch structures.

The RGB-only model mainly attended to visible colour and texture regions on the bunch surface. However, the attention regions appeared relatively scattered in some samples, suggesting that RGB information alone may not always provide sufficiently distinct features for separating adjacent ripeness stages. The CIEL*a*b*-only model showed attention patterns more strongly associated with colour-discriminative regions, suggesting that transformation from RGB to CIEL*a*b* helped emphasize chromatic differences related to ripeness.

For the RGB + IR model, the activation maps indicated that IR information altered the model focus regions. The model attended not only to visible fruitlet colour areas but also to regions associated with differences in IR reflectance. The CIEL*a*b* + IR model produced the most distinct activation over FFB regions, particularly fruitlet areas associated with ripeness-related colour and reflectance changes. This observation is consistent with the quantitative results, in which the CIEL*a*b* + IR model achieved the highest weighted-average precision, recall, and F1-score.

Neither the Color and IR images nor the CNN class activation heatmaps reveal significant differences between correctly classified and misclassified samples; both groups generally follow the trends shown in Figures 4 and 5.

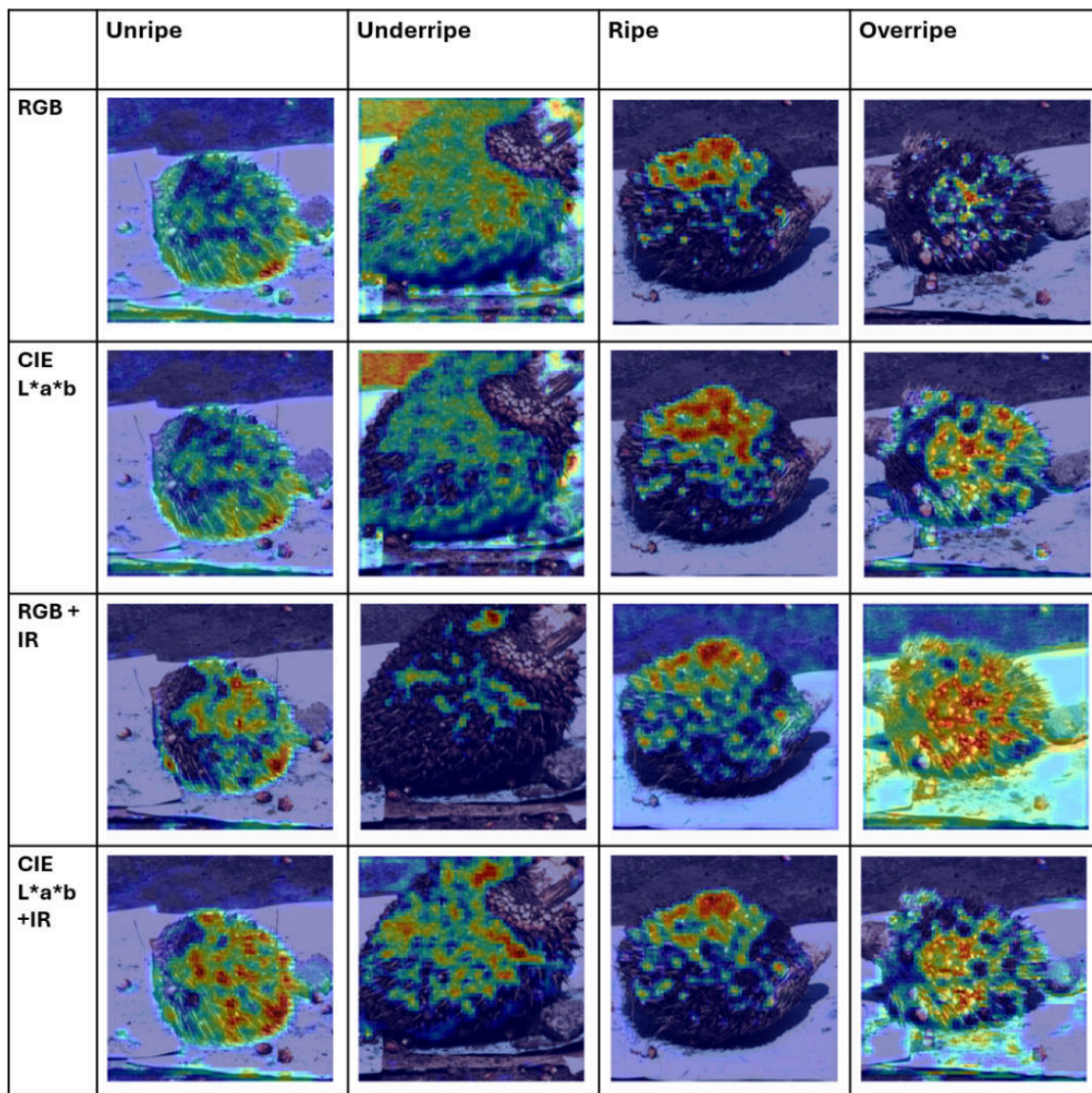


Figure 5. Class activation heat maps of CNN models trained using different input image formats for FFB ripeness classification. Columns show unripe, underripe, ripe, and overripe samples from left to right; rows represent RGB, CIE L*a*b, RGB + IR, and CIE L*a*b + IR models. Warmer colours indicate stronger contribution to the classification decision.

3.4. Classification Performance

Table 2 compares the performance of the four CNN-based classification models. Because all models used the same CNN architecture, differences in performance are attributed mainly to image input representation and the inclusion of IR information. The RGB-only model achieved weighted-average precision, recall, and F1-score values of 0.79, 0.76, and 0.76, respectively. Adding IR to RGB increased the corresponding values to 0.82, 0.80, and 0.80, indicating that IR information provided complementary features that improved ripeness discrimination. The classification results are consistent with the spectral observations shown in Figure 3, where the IR-sensitive wavelength region exhibited clear reflectance differences among ripeness stages.

Table 2. Comparative classification performance of CNN models trained using different image input formats for FFB ripeness classification. Precision, recall, and F1-score with the respective standard deviation ($\pm$ s.d) are reported for each ripeness class, and weighted-average values summarize the overall performance of each input format.

Input Format	Class	Precision	Recall	F1-Score
RGB-only	Overripe	0.92 ± 0.11	0.67 ± 0.05	0.77 ± 0.04
	Ripe	0.75 ± 0.02	0.87 ± 0.09	0.80 ± 0.04
	Underripe	0.66 ± 0.07	0.83 ± 0.10	0.73 ± 0.05
	Unripe	0.89 ± 0.06	0.64 ± 0.14	0.73 ± 0.09
	Weighted average	0.79 ± 0.03	0.76 ± 0.04	0.76 ± 0.04
RGB + IR	Overripe	0.89 ± 0.08	0.92 ± 0.10	0.90 ± 0.06
	Ripe	0.86 ± 0.08	0.79 ± 0.08	0.82 ± 0.04
	Underripe	0.68 ± 0.06	0.83 ± 0.08	0.74 ± 0.04
	Unripe	0.90 ± 0.06	0.69 ± 0.11	0.77 ± 0.06
	Weighted average	0.82 ± 0.03	0.80 ± 0.03	0.80 ± 0.04
CIEL*a*b*-only	Overripe	0.93 ± 0.10	0.68 ± 0.06	0.78 ± 0.06
	Ripe	0.78 ± 0.04	0.89 ± 0.10	0.82 ± 0.04
	Underripe	0.65 ± 0.05	0.82 ± 0.08	0.72 ± 0.04
	Unripe	0.88 ± 0.06	0.66 ± 0.10	0.75 ± 0.06
	Weighted average	0.79 ± 0.03	0.77 ± 0.03	0.76 ± 0.03
CIEL*a*b* + IR	Overripe	0.95 ± 0.05	0.95 ± 0.09	0.95 ± 0.05
	Ripe	0.91 ± 0.08	0.78 ± 0.02	0.84 ± 0.04
	Underripe	0.66 ± 0.03	0.83 ± 0.06	0.74 ± 0.03
	Unripe	0.89 ± 0.08	0.72 ± 0.08	0.79 ± 0.02
	Weighted average	0.84 ± 0.02	0.81 ± 0.02	0.81 ± 0.02

The best overall performance was obtained by the CIEL*a*b* + IR model, which achieved weighted-average precision, recall, and F1-score values of 0.84, 0.81, and 0.81 respectively. Compared with RGB-only, the CIEL*a*b* + IR model improved weighted-average precision by 0.05, recall by 0.05, and F1-score by 0.05. These results indicate that CIEL*a*b* colour representation and IR information provide complementary input features for CNN-based FFB ripeness classification.

In addition to achieving the highest weighted-average precision, recall, and F1-score, the proposed CIELab + IR model also exhibited the lowest standard deviations across all three evaluation metrics. The weighted-average standard deviations for precision, recall, and F1-score were ± 0.02 , ± 0.02 , and ± 0.02 , respectively, compared with ± 0.03 – 0.04 for the RGB-only model. The lower standard deviations indicate that the proposed CIEL*a*b* + IR model delivered more consistent performance across different random train–test splits and was less affected by variations in the training data. This suggests that combining CIEL*a*b* colour information with IR reflectance produced a more robust feature representation, enabling the CNN to distinguish ripeness classes more consistently with clearer decision boundaries. As a result, the model produced more reliable and repeatable predictions than the other input representations. In contrast, the larger standard deviations observed for the other models indicate greater variability in classification performance across different data partitions, suggesting that their extracted features were less effective in separating neighbouring ripeness classes.

At the class level, IR information substantially improved overripe classification. Overripe recall increased from 0.67 for RGB-only to 0.92 for RGB + IR and 0.95 for CIEL*a*b* + IR, while the corresponding F1-score increased from 0.77 to 0.90 and 0.95. As the fruit becomes overripe, physical changes such as chlorophyll degradation, moisture variation, and structural changes become more pronounced.

The ripe and unripe classes also benefited from the improved input representations. For the ripe class, precision increased from 0.75 for RGB-only to 0.91 for CIEL*a*b* + IR, although recall decreased from 0.87 to 0.78. This indicates that the CIEL*a*b* + IR model was

more conservative in predicting the ripe class, reducing false-positive ripe classifications while missing some true ripe samples. For the unripe class, F1-score improved from 0.73 for RGB-only to 0.79 for CIEL*a*b* + IR, showing improved balance between precision and recall.

The underripe class remained the most challenging category across all models, primarily due to its high rate of false positives. Underripe precision was consistently lower than that of the other classes, ranging from 0.65 to 0.68. This is consistent with the transitional nature of underripe FFB, which lies between unripe and ripe conditions. Fruitlets within the same bunch may show uneven ripeness development, resulting in mixed visual and spectral characteristics. Consequently, underripe bunches may share features with both unripe and ripe classes, making them more difficult to classify precisely.

Although underripe recall was relatively high across models, ranging from 0.82 to 0.83, the lower precision indicates that some samples from other ripeness classes were incorrectly predicted as underripe. This suggests that the models detected many true underripe samples but had difficulty defining a precise decision boundary for this transitional class.

Overall, the results confirm that both IR information and CIEL*a*b* colour representation affect CNN classification performance. CIEL*a*b* alone produced only a small improvement over RGB alone, whereas the greatest performance gain was achieved when CIEL*a*b* was combined with IR information. These findings support the proposed four-channel CIEL*a*b* + IR image input as the best-performing solution among the evaluated input formats for CNN-based FFB ripeness classification. Nevertheless, underripe FFB remains a major classification challenge because of its transitional and heterogeneous ripeness characteristics.

3.5. Normalized Confusion Matrix Analysis and Classification Error Patterns

Figure 6 shows Normalized Confusion Matrices of average counts for RGB, RGB+IR, CIE L*a*b* and CIE L*a*b*+IR input. To further investigate category-specific decision boundaries and misclassification patterns across input representations, normalized confusion matrices aggregated across the evaluation runs were analyzed for the four model variants (RGB, RGB + IR, CIEL*a*b*, and CIEL*a*b* + IR). The most prominent diagnostic improvement observed upon integrating the IR channel lies in resolving the severe boundary confusion between the overripe and ripe classes. For 3-channel color representations (RGB and CIEL*a*b*), true overripe samples suffered from high misclassification rates, with 28.00% and 23.00% falsely predicted as Ripe, respectively. Consequently, baseline RGB achieved a correct classification rate of only 67.00% for overripe bunches. Upon incorporating the IR image, classification accuracy for overripe samples surged dramatically to 92.00% in the RGB + IR model and reached a peak of 95.00% in the proposed CIEL*a*b* + IR architecture. This sharp reduction in false-positive Ripe predictions indicates that cellular structural changes and moisture/lipid variations captured within the 700–1100 nm IR range supply essential discriminatory features that visible surface color alone fails to convey under dynamic outdoor sunlight.

Across all four input representations, the primary remaining source of classification error consistently occurs at the transition boundary between unripe and underripe fresh fruit bunches (FFBs). For baseline RGB, 35.62% of true Unripe samples were misclassified as underripe, yielding an unripe accuracy of 64.38%. Transforming the color space to CIEL*a*b* marginally improved Unripe accuracy to 66.25%, with 33.75% misclassified as underripe. The inclusion of the IR channel systematically suppressed this transition error down to 28.12% in the CIEL*a*b* + IR representation, pushing unripe classification accuracy to its highest level of 71.88%. This residual confusion between unripe and underripe stages reflects the continuous nature of fruitlet maturation, where surface chlorophyll

degradation occurs gradually, causing adjacent class overlap along the decision boundary during early maturation phases.

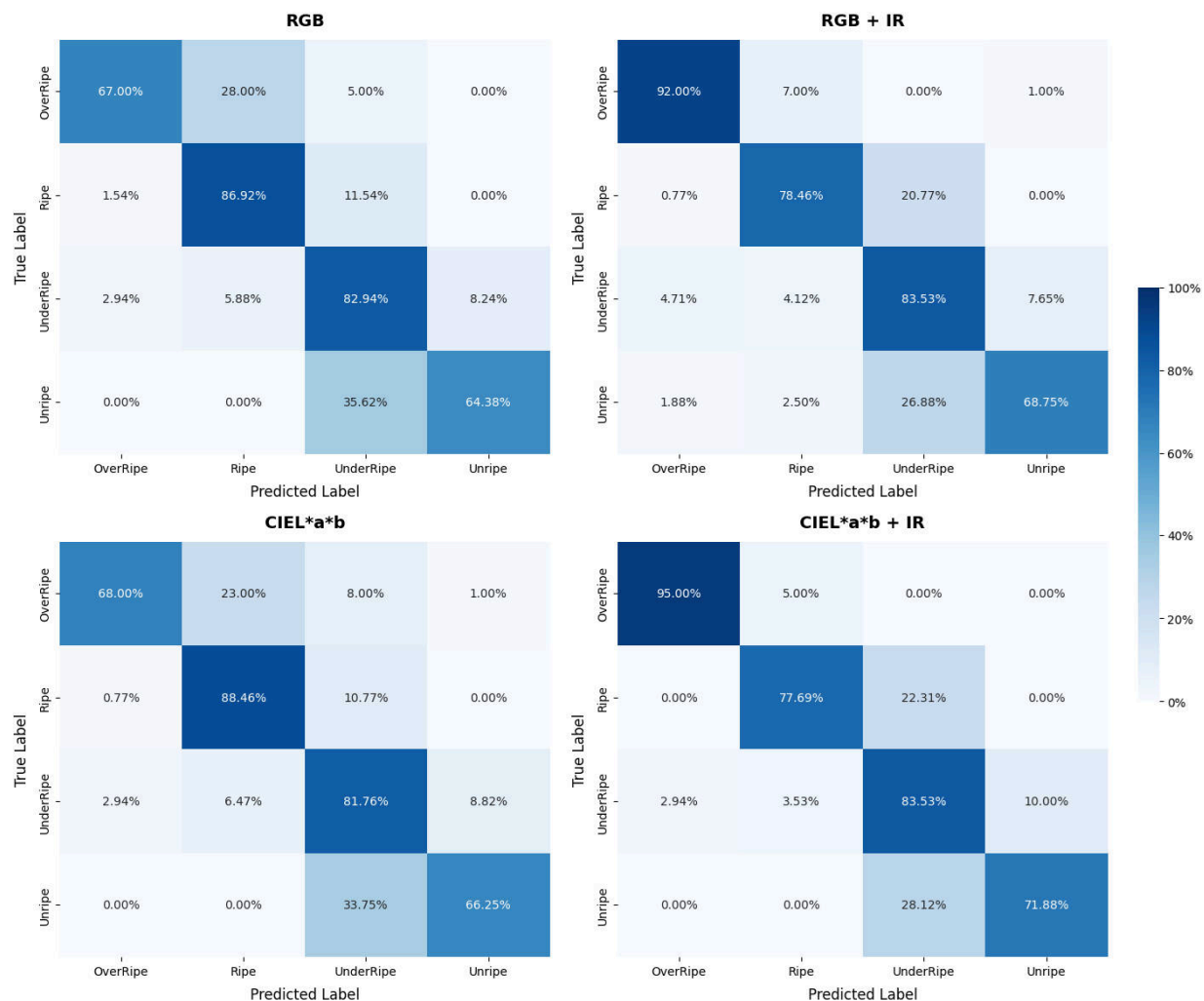


Figure 6. Normalized Confusion Matrices for RGB, RGB + IR, CIE L*a*b* and CIE L*a*b* + IR input.

Overall, the proposed 4-channel CIE L*a*b* + IR input representation demonstrated the most balanced and robust diagonal confusion profile across all four ripeness stages. Specifically, the model achieved peak accuracy across all variants for the overripe (95.00%) and Unripe (71.88%) classes, while maintaining strong performance for Ripe (77.69%) and underripe (83.53%) bunches. Crucially, for the Ripe class, the CIE L*a*b* + IR model eliminated all false-positive errors into the non-adjacent overripe and unripe categories (0.00%), restricting all residual uncertainty strictly to the neighbouring underripe class (22.31%). In summary, while color space transformation alone (CIE L*a*b*) offers minimal gain over standard RGB, concatenating CIE L*a*b* chromatic features with IR image substantially mitigates major misclassifications—particularly for overripe bunches and optimizes decision boundary stability across adjacent maturation stages.

3.6. Statistically Significant Analysis

To determine whether the observed performance improvement of the proposed CIE L*a*b* + IR representation over the baseline RGB model is statistically meaningful, a paired statistical significance test was conducted across the 10 randomized evaluation splits. Because both model variants were evaluated on the exact same test splits in each run, the weighted average Precision scores yielded 10 naturally paired observations ($n = 10$). A

Shapiro–Wilk test performed on the paired differences ($\text{CIEL}^*a^*b^* + \text{IR} - \text{RGB}$) confirmed a normal distribution ($W = 0.9471, p_{SW} = 0.6349 > 0.05$), where W is Shapiro–Wilk test statistic and p_{SW} is Shapiro–Wilk p -value. Consequently, a two-tailed paired t -test was executed, revealing a statistically significant improvement in weighted average precision for the $\text{CIEL}^*a^*b^* + \text{IR}$ model ($t(9) = 2.5634, p = 0.0305 < 0.05$), with a mean positive difference of 0.0420 ± 0.0518 (+4.2). This confirms that the gain achieved by integrating near-infrared images is statistically significant and outweighs evaluation variance across random dataset splits.

3.7. Computational Efficiency and Training Duration

To evaluate the computational overhead of different models, the average training execution time across the 10 randomized split runs was recorded for each model representation. The baseline RGB model required an average training time of 52.8 ± 13.7 min. In comparison, transforming the color space to $\text{CIEL}^*a^*b^*$ reduced the average training duration to 42.4 ± 5.5 min. Interestingly, integrating the single-channel Near-Infrared (IR) reflectance layer resulted in even faster convergence across training epochs: the $\text{RGB} + \text{IR}$ model averaged 38.8 ± 7.6 min, while the proposed $\text{CIEL}^*a^*b^* + \text{IR}$ representation achieved the shortest overall training duration at 35.3 ± 6.3 min. Notably, the addition of a 4th input channel did not impose an overall computational time penalty. Rather, this empirical reduction in training duration suggests that incorporating IR information may enhance feature discriminability at the input layer, potentially aiding optimization and allowing the network to reach convergence in fewer epochs. Overall, the proposed $\text{CIEL}^*a^*b^* + \text{IR}$ model achieved a 33.1% reduction in average training time compared to the baseline RGB model, alongside improved execution stability as evidenced by the reduced standard deviation across split runs.

3.8. Limitations and Future Work

While this study demonstrates a successful proof-of-concept for 4-channel input fusion in oil palm fruit ripeness classification, several practical constraints and limitations should be considered. First, the primary constraint of this study stems from dataset volume and environmental scope. The current evaluation was conducted on a dataset of 279 paired image sets collected from a localized plantation site. While repeated stratified/random 80:20 train-test split stratification and data augmentation were implemented to prevent overfitting and assess statistical stability across dataset splits, a substantially larger dataset encompassing diverse oil palm cultivars, multiple harvesting locations, and varying seasonal conditions is necessary to establish broader operational generalizability.

Furthermore, the experimental setup evaluated single static FFB against relatively controlled background conditions under natural afternoon sunlight. In real-world plantation environments, automated grading platforms must contend with multi-objective concurrency, dynamic leaf canopy shading, harsh sunlight glare, and severe physical fruitlet occlusion. Additionally, from a hardware deployment perspective, the current dual-camera setup relies on two separate physical sensors positioned side-by-side. Maintaining channel pixel alignment across varying working distances requires manual calibration and static capture protocols. Translating this methodology to mobile field-grading devices or autonomous harvesting machinery will necessitate compact multispectral sensor integration, optical co-axial beam-splitters, or automated real-time image registration algorithms to resolve dynamic parallax.

To address these limitations, future research will focus on expanding data acquisition across dynamic plantation environments and unconstrained field conditions. Upcoming work will benchmark the lightweight custom network against modern Vision Transform-

ers (ViT) and state-of-the-art real-time object detection frameworks (such as Ultralytics YOLOv26 and RF-DETR) to evaluate multi-target detection capabilities. Finally, future iterations will explore the direct integration of quantitative physicochemical assays, such as mesocarp oil content extraction, to further refine quality assessment and facilitate end-to-end integration with automated harvesting systems.

4. Conclusions

This study evaluated RGB, RGB + IR, CIEL*a*b*, and CIEL*a*b* + IR image representations for CNN-based FFB ripeness classification using a common model architecture. The CIEL*a*b* + IR model achieved the best overall performance, confirming that the proposed four-channel CIEL*a*b*+ IR representation is the most effective input format among the evaluated models. The improvement is attributed to the complementary roles of CIEL*a*b* colour-space transformation and IR reflectance imaging: CIEL*a*b* provides a more discriminative colour representation with reduced dependence on brightness, while IR reflectance captures ripeness-related physical changes that are not fully represented by visible colour information alone. Although the proposed representation improved overall classification performance, underripe FFB remained the most challenging class, where it produced the highest rate of false positives because of its transitional and heterogeneous ripeness characteristics.

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